

Artist Friendly Physically Plausible Attenuation

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Figure 1: Modifying a subsurface scattering material to produce a warmer transmission using attenuation versus our constrained method. The center Stanford dragon exhibits undesirable green scattering and unnatural gradients while our method on the right guarantees a plausible result.

ABSTRACT

In this paper we explore the intricate relationship between volumetric albedo and attenuation. When rendering realistic skin and complex FX elements, the traditional approach of authoring absorption and scattering can pose a challenge when trying to achieve a desired opacity, brightness and saturation. The alternate parameterisation of albedo and attenuation can also produce unpredictable results. Focusing on Mie and Rayleigh scattering, we develop three novel techniques that derive physically plausible attenuation values from albedo. By understanding the implications of Rayleigh scattering, and adjusting albedo in an intuitive manner, artists can achieve predictable and consistent results with volumetric rendering workflows.

CCS CONCEPTS

• Computing methodologies → Reflectance modeling.

KEYWORDS

VFX, volumetric rendering, subsurface scattering

ACM Reference Format:

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1 INTRODUCTION

In the ever-evolving landscape of computer graphics, volumetric rendering has become an integral part of modern production pipelines. It is responsible for creating realistic and visually compelling representations of mediums such as smoke, atmospheres, skin, and other complex materials. As the industry continues to push the boundaries of realism and artistic expression, the development of novel techniques is important.

In recent years, a clear trend has emerged in the field of computer graphics, with a strong emphasis on adopting physically based approaches to rendering. These approaches aim to simulate the behavior of light and materials in a manner consistent with real-world physics, ultimately resulting in more accurate and believable imagery.

Recognizing the need for a balance between physical accuracy and artist-friendly workflows, researchers and developers have focused their efforts on creating techniques that cater to both objectives. The idea is to develop methods that not only adhere to the underlying principles but also provide artists with the flexibility and control they need to create high-quality imagery while maintaining a strong foundation in physical accuracy.

2 BACKGROUND

In recent years, volumetric rendering techniques have experienced significant advancements, particularly in the field of subsurface

scattering. Since 2016, volumetric methods for subsurface scattering, also called path-traced or random-walk, have been extensively utilized to achieve highly realistic and visually appealing results[Burley 2015, sec. 2.5][Matt Jen-Yuan Chiang and Burley 2016]. Studios like Framestore have adopted albedo/attenuation-based parameterization, aligning the authoring process with previously established workflows[Jensen and Buhler 2002][Matt Jen-Yuan Chiang and Burley 2016][Walster 2017][Georgiev et al. 2023]. These parameters allow artists to express surface appearances as the color it reflects and the distance light travels through it, respectively.

However, despite the improvements in shading and parameterization, we found the asset authoring process could become less predictable and, in some instances, more challenging. Artists often grapple with the complexities associated with managing the intricate relationships between albedo and attenuation, especially when attempting to reproduce skin appearances from reference imagery. Attenuation not only controls the distance light transmits through matter but also can under the right conditions inadvertently affect the color of reflected light. Intricate FX elements such as smoke and explosions can also face their own set of challenges. Consequently, further research and development were necessary to streamline the asset authoring process while maintaining the practicality offered by albedo-based parameterization.

3 MOTIVATION

Initially, our objective was to generate theoretical albedo and attenuation values for skin by utilising absorption and scattering measurements from its various components, aiming to create realistic skins using a dual-layer shading model. Our intention was to investigate the effects of varying concentrations of epidermal melanin/beta-carotene and dermal haemoglobin/bilirubin[Jacques 2015][Leeuwen and Baranoski 2018]. We sought a reliable skin texturing method to address dynamic skin effects such as bloodflow. It proved to be challenging in lookdev as our typical approach relies on reference imagery from which we would now need to extract a range of concentration levels.

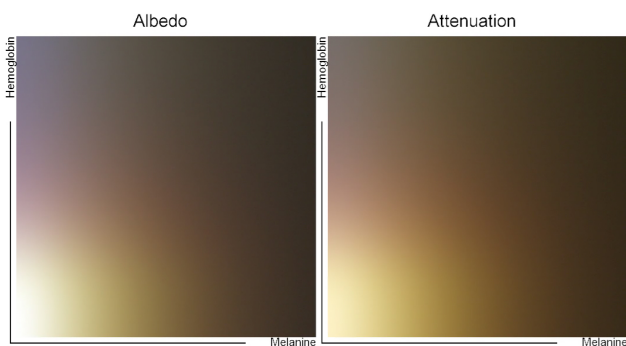


Figure 2: Single-scattering albedo and attenuation of skin for ranges of eumelanin and hemoglobin concentrations

Through intuition, we observed a relationship between the single-scattering albedos and attenuations produced (Figure 2) and wondered if we could discern their interdependence. We noticed that their relationship’s complexity is primarily determined by the type

of scattering involved (Figure 4,5). In this paper we look into establishing generalised workflows beyond our original scope of skin authoring only.

4 AUTHORIZING

In traditional volumetric rendering, defining absorption and scattering coefficients can be unintuitive, making it difficult to achieve the desired opacity, brightness and saturation. Albedo parameterization is generally considered more efficient for artists in texturing workflows when used for subsurface scattering[Burley 2015, sec. 2.4 para. 2].

4.1 Subsurface scattering

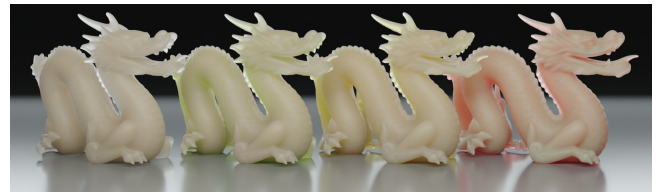


Figure 3: Subsurface scattering materials using unconstrained attenuation values producing implausible appearances

Switching to an artist-friendly albedo/attenuation parameterization was promising but we noticed that it yielded some inconsistent results in production. Attenuation maps started becoming a challenge when deviating from diffusion models, as volumetric rendering is more sensitive. Albedo/attenuation imbalances led to unnatural scattering coefficients.

Simple ad-hoc solutions had been in use by artists since the use of diffusion models. However, these solutions offered no guarantee of consistency using volumetric rendering, and their effectiveness tended to diminish as the authoring process became more complex. In some cases, further adjustments would only exacerbate the issue, highlighting the need for more refined and predictable methods.

4.2 FX

The accurate authoring of intricate FX elements, such as smoke and explosions, presents a challenge as well. Smoke appearances can range from bright pyrolysis to dark soot clouds through partial combustion. When modifying scattering and absorption properties to represent these various states, the opacity of the volume can be unexpectedly affected, contrary to artists’ expectations of relying solely on density grids. Addressing this issue within the context of a gradient between different states of smoke can be un-intuitive.

Albedo/Attenuation parameterization allows the simplification of this process by normalizing attenuation, thereby offering more predictability. However, when adjustments are made to albedo brightness, some intricacies may arise, which are usually not considered.

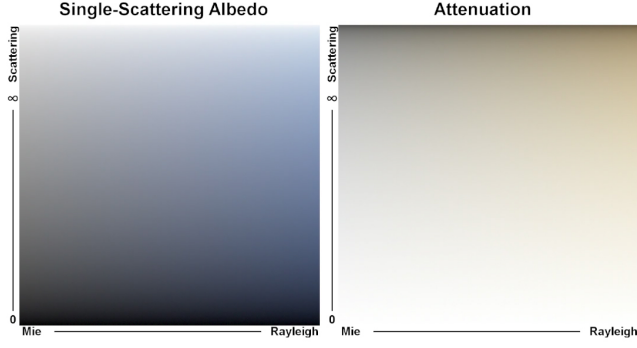


Figure 4: The single-scattering albedo and attenuation values for a range of scattering coefficients between Mie and Rayleigh in the absence of absorption

5 PARAMETER SPACE

5.1 The scattering space

Natural scattering phenomena have inherent limitations, mainly influenced by particle size and indices of refraction. We focus on Mie, Rayleigh and Tyndall scattering, which cover most common scattering behaviours in the visible light spectrum.

Rayleigh scattering describes light scattering by particles much smaller than the wavelengths of the incident light, with shorter wavelengths (e.g., blue light) scattering more effectively than longer wavelengths (e.g., red light). Mie scattering creates angular-dependent color variations in single-scattering events (e.g., rainbows) but averages out in multiple scattering situations giving a mostly monochromatic appearance. Tyndall scattering is observed in colloidal suspensions, where light interacts with particles comparable to or larger than its wavelength. This results in scattered light, similar to Rayleigh scattering. Therefore, we have decided to omit its discussion from this paper, working under the assumption that its coefficients are adequately represented by Rayleigh scattering.

Media made up of different-sized particles can lead to a mixture of Mie and Rayleigh scattering. Figures 4 and 5 illustrate how the albedo and attenuation change across the gradient between the two. They also display all the plausible values that could occur in the presence of monochromatic absorption. Any combinations beyond what is shown in them would exceed what Mie and Rayleigh scattering can represent. In other words, they would be considered implausible.

With Mie scattering, the relationship is simple while the one for Rayleigh scattering is more intricate. Its albedo exhibit a prominent blue hue with darker values in contrast to its attenuation that instead shifts towards a distinct orange color as the albedo brightens. This difference in behaviour makes the authoring process difficult to achieve consistent plausible results.

As a result, a transition from black to white, which could be expressed by scalar values, now requires representation through two color values in the presence of Rayleigh scattering, substantially increasing the memory and storage requirements particularly in the case of FX elements.

We believe the natural constraints of scattering justify limiting the coefficients to a space delimited by the Rayleigh and Mie scattering coefficients to improve authoring workflows.

5.2 The absorption space

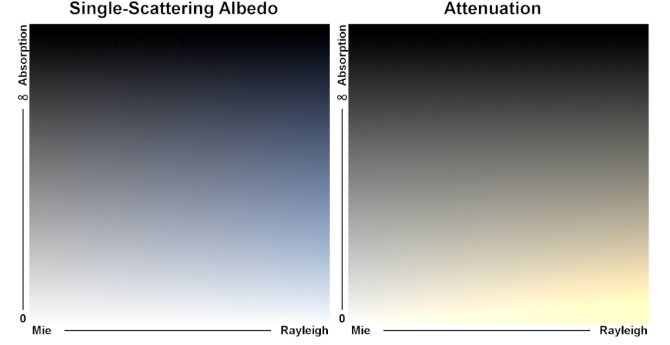


Figure 5: The single-scattering albedo and attenuation values for a range of monochromatic absorption coefficients between Mie and Rayleigh using an arbitrary scattering amount

The absorption component is entirely dependent on the molecular composition of the medium. Each element and molecule possesses a distinct spectral absorption footprint, which is a result of the interplay between quantum physics and chemistry.

Due to the wide range of possible compositions, the absorption component cannot be constrained, guaranteeing the ability to recreate any potential medium.

6 SIMPLIFYING THE WORKFLOW

First lets consider the following definitions of single-scattering albedo α and attenuation RGB colors [Walster 2017] where RGB quantities are expressed as boldface and scalar as italic:

$$\sigma_t = \sigma_a + \sigma_s \quad (1)$$

$$\text{attenuation} = \frac{1}{\sigma_t} \quad (2)$$

$$\alpha = \sigma_s \text{attenuation} \quad (3)$$

The attenuation is in theory readily available. Simplifying the authoring process to leverage the information already encoded in the albedo could ensure more predictable volumetric renders. If we could constrain one of the albedo components, we would be able to derive the rest. Fortunately, as previously discussed, the scattering space is limited, which significantly simplifies the problem.

Inherently, albedo is a unitless measure of the fraction of reflected light off a surface. It indirectly encodes the ratio between scattering and absorption, eliminating the need for real-world coefficients. Consequently, we can make several assumptions. We discuss two techniques for deriving attenuation based on different assumptions about the scattering and absorption properties and a third to address some intricacies of FX volumes authoring.

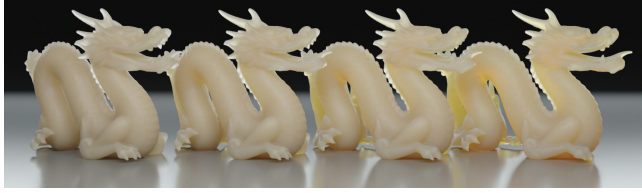


Figure 6: Constant Scattering. Increasing Rayleigh contribution from left to right.

Constant Scattering. This method is particularly useful for skin rendering. The scattering is simply defined as the proportion between Rayleigh and Mie scattering to reduce the range of possible attenuation values. As a result, albedos drive the absorption composition where lower values automatically lead to a more opaque appearance. Figure 6, 12

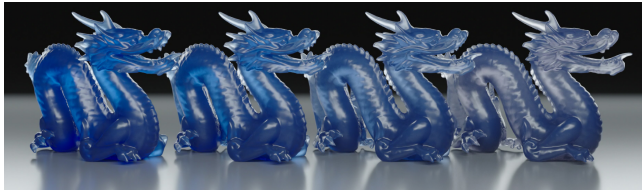


Figure 7: Constant Absorption. Increasing Rayleigh contribution from left to right.

Constant Absorption. This method is well-suited for rendering materials such as ice, water and crystals. By extracting the scattering component and constraining it to a coefficient with a user-defined Rayleigh contribution we can then extract the absorption.

Consequently, albedo brightness controls the scattering intensity where brighter values lead to a more opaque appearance, providing a more realistic rendering of these types of materials. Figure 7, 13



Figure 8: Rayleigh Injection. Increasing Rayleigh contribution from left to right.

Rayleigh Injection. This approach has been developed primarily for rendering FX volumes. We enforce normalisation of attenuation, ensuring that the opacity control is governed solely by density values.

Given a float or color albedo, we assume the monochromatic Mie as the base scattering. Subsequently, we weigh in Rayleigh scattering coefficients, leading to a modified albedo and associated attenuation.

This method not only streamlines the workflow but also enables the generation of more accurate visual effects, without the need to explicitly account for the complex behaviour introduced by Rayleigh scattering in the dataset. Figure 8, 14

7 IMPLEMENTATION

7.1 Attenuation derivation with constant scattering

We initially derive the approximate single-scattering albedo α from the multiple scattering albedo A by adopting Disney’s approach [Matt Jen-Yuan Chiang and Burley 2016].

$$\alpha = 1 - e^{-5.09406A + 2.61188A^2 - 4.31805A^3} \quad (4)$$

Following this, we use w to weigh between Mie m (white) and Rayleigh r (blue) scattering coefficients to produce the coefficient s as shown on the left side of Figure 9. We assign a value of (0.553, 0.808, 1.64) to r , for details refer to the Appendix A.

$$s = rw + m(1 - w) \quad (5)$$

We then divide the single-scattering albedo with it, yielding a plausible attenuation value that can be further scaled using a scalar density parameter.

$$\text{attenuation} = \frac{\alpha}{s} \quad (6)$$

The primary advantage of this workflow lies in the reduction of the attenuation color parameter down to the scalar value w , which effectively constrains the resulting scattering coefficients within the bounds of Mie and Rayleigh scattering. Users don’t need to worry about the albedo/attenuation relationship. It ultimately leads to more predictable and consistent outcomes in our authoring process.

7.2 Attenuation derivation with constant absorption

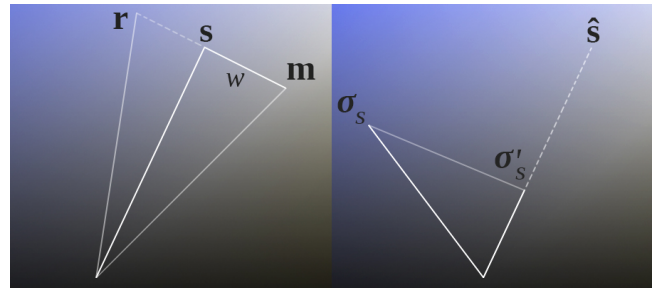


Figure 9: On the left we show the user-defined w to weigh between the Rayleigh and Mie coefficients giving s . On the right we show the projection of the isolated scattering σ_s from the albedo onto the target coefficient producing a new constrained coefficient σ_{s1} .

We begin by estimating the single-scattering albedo using equation 4. Next, make the assumption that absorption is white and

isolate scattering from the albedo. Important to note that every channel of α should be clamped below 1 to prevent divisions by 0.

$$\sigma_s = \frac{\alpha}{1 - \alpha} \quad (7)$$

We then project the scattering coefficient σ_s onto the normalized vector $\hat{s}$ defined with equation 5 to get a new constrained scattering σ'_s as shown in Figure 9.

$$\sigma'_s = \hat{s}(\hat{s} \cdot \sigma_s) \quad (8)$$

With σ'_s we can now isolate for attenuation.

$$\text{attenuation} = \frac{\alpha}{\sigma'_s} \quad (9)$$

The Rayleigh weight parameter enables us to control how much blue from the albedo is utilized in scattering as opposed to absorption.

7.3 Attenuation derivation with Rayleigh Injection

In this approach, the single-scattering albedo is either explicitly defined or estimated depending on the user's workflow preference. We then derive the absorption in a manner that maintains attenuation, assuming a white scattering coefficient. Here we have to clamp α above 0 to prevent divisions by 0.

$$\sigma_a = \frac{1}{\alpha} - 1 \quad (10)$$

Subsequently, we reconstruct the single-scattering albedo and attenuation by incorporating the weighted scattering coefficient s using equation 5 and the derived absorption σ_a .

$$\alpha' = \frac{s}{\sigma_a + s} \quad (11)$$

$$\text{attenuation} = \frac{1}{\sigma_a + s} \quad (12)$$

And finally we normalize the attenuation using the average of all channels so that the opacity is entirely driven by density.

The resulting albedo differs from the initial value, but this method significantly streamlines the workflow for creating the intricate appearance of smoke. The primary consideration for the artist at this stage is to grossly estimate the average particle size of the FX element, which is used to drive the Rayleigh weight.



Figure 10: Wedging Rayleigh weight between the 3 left renders. The render on the right shows a complete explosion using the remapped albedo to drive Rayleigh weight.

The albedo's brightness can sometimes serve this purpose in an ad-hoc manner (Figure 10), as particles with high absorption

values at the scale of Rayleigh scattering are unlikely to occur. This can lead to interesting memory/storage optimisations by avoiding using 2 color grids where a single scalar grid might achieve the same result.

This approach streamlines the process of generating various templates for FX types as well. For instance, the main difference between clouds/mist/steam and pyrolysis is ultimately the Rayleigh weight.

8 NEW WORKFLOW

As part of our workflow, we have found it useful for artists to gain a better understanding of the implications of Rayleigh scattering. By carefully examining reference images for color dissociations, they can identify cues for the presence of Rayleigh scattering. Knowledge of the proportions between Mie and Rayleigh scattering for different materials also allows artists to achieve their desired appearance by adjusting the albedo accordingly.

8.1 Attenuation



Figure 11: Albedo change of ears of a character showing the effect on transmission.

In contrast to traditional approaches, changes in transmitted color should now be authored directly from albedo if the Rayleigh contribution is known ($w \approx 0.64$ for skin [Jacques 2015]). As an example, when using the constant scattering method, red transmission through ears can be driven with a more saturated red albedo as shown in Figure 11, rather than attenuation. Introducing darker values automatically gives a more opaque appearance for darker skin tones, veins, moles, or freckles. Inversely, when using the constant absorption method, using white values to represent the internal reflections of cracks and air bubbles in ice will lead to a more opaque result.

These workflows achieve the desired goal of preventing color inversions to the diffusion. Attempting to push attenuation beyond the capabilities of these techniques will lead to undesirable scattering coefficients, emphasizing the importance of these approaches in achieving predictable and realistic results in volumetric rendering.

8.2 Phase function

With our new ability to define a Rayleigh weight, we could potentially drive the phase function. This relationship would allow us to manage the scattering behavior of light more effectively. Interpolating between forward scattering and an isotropic lobe could be linked to the Rayleigh weight; however, particle size distributions have a greater range of effect on the Mie scattering profiles than what can be expressed in albedos alone. Furthermore, while Tyndall

scattering exhibits similar colors to Rayleigh scattering, its scattering profile aligns more closely with Mie scattering. Instead, we leave the control to the users to approximate the combined phase function of Mie and Rayleigh scattering with Henyey-Greenstein lobes [Henyey and Greenstein 1940].

9 LIMITATIONS

Although we feel that the proposed parameterizations cover many typical lookdev scenarios, they are not a comprehensive solution. In complex layered volumetric regions, such as eyes, darker albedo values can't be used to simultaneously represent dense veins and a transparent cornea. In these situations we continue to describe macro features using density variations.

Additionally, the intricate interplay of scattering and absorption within volumetric materials necessitates multiple scattering events to accurately capture a materials appearance. In many situations this can be computationally expensive, posing a challenge for artists and rendering engineers needing to balance realism and physical plausibility with computational efficiency.

10 FUTURE WORK

In future work, we would like to investigate methods to constrain the Rayleigh contribution in mediums with lower albedo. In some situations we expect the probability of Rayleigh scattering to diminish in the context of increased absorption values, as these typically imply larger particles. We would also like to investigate further how it might be able to drive phase functions in production contexts. It would also be important to do a more thorough investigation of Tyndall scattering and it's relationship to the Mie/Rayleigh space presented. By refining our understanding of all these relationships, we may improve our rendering capabilities for mediums such as smoke.

Additionally, we plan to explore the potential role of Rayleigh scattering in hair and fur rendering. Understanding the impact of this scattering mechanism on its appearance could lead to more realistic representations and further enhance the visual quality of some exotic creatures.

11 CONCLUSION

In conclusion, we have explored the relationship between albedo and attenuation, and introduced alternate parameterizations that aim to improve authoring consistency and robustness. Our approach, based on Mie and Rayleigh scattering, constrains the user parameter space and derives plausible attenuation values from albedo and the new Rayleigh weight parameter. We have found these techniques have enabled our artists to achieve a visually accurate appearance across various media, such as smoke, skin, and teeth, accelerating our look development process, and ensuring cohesion between any phenomena that exhibits Rayleigh scattering.

ACKNOWLEDGMENTS

Stanford Dragon model provided by the Stanford 3D Scanning Repository. Thanks to Manuel Gamito and the Framestore rendering team for valuable insights.

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Coefficients. We define the Mie coefficient $\mathbf{m}$ as white (1.0, 1.0, 1.0), fully reflective of all wavelengths.

We define the Rayleigh coefficient $\mathbf{r}$ with equation 13 [Frisvad 2011] by choosing the dominant wavelengths related to your RGB color primaries [Schanda 2007] or by spectrally integrating it to your working colorspace. We then normalize the coefficient using the average of all channels.

$$\mathbf{r} = \frac{1}{\lambda^4} \quad (13)$$

We get an approximate value for $\mathbf{r}$ of (0.553, 0.808, 1.64) in sRGB and (0.564, 0.747, 1.688) in ACEScg.

The Mie to Rayleigh space. We show how to reproduce figures 4 and 5 in sRGB.

Figure 4: The x axis is a linear gradient between the $\mathbf{m}$ and $\mathbf{r}$ colors which gets multiplied by a linear 0-1 gradient in the y axis using the mapping $\frac{1}{y^{0.4545}-1}$ defining σ_s . We assume σ_a is white and reconstruct α and **attenuation** using equation 1,2,3.

Figure 5: The x axis is a linear gradient between the $\mathbf{m}$ and $\mathbf{r}$ colors defining σ_s . The y axis defines a linear 0-1 gradient for absorption coefficients using the mapping $\frac{1}{y^{2.2}-1}$ defining σ_a . We then reconstruct α and **attenuation** from these coefficient gradients using equation 1,2,3.

Wedges. The wedges in figures 12, 13, 14 show the effect of albedo on opacity using $w = 0.67$.



Figure 12: Constant Scattering.

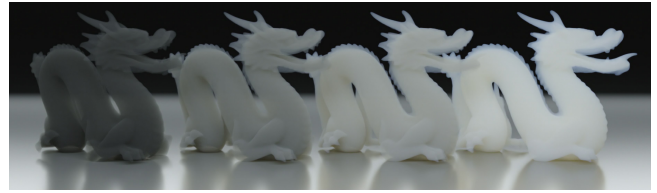


Figure 14: Rayleigh Injection.



Figure 13: Constant Absorption.